Story Reel: Chaptered Memory Albums from a Family NAS Photo Library

UGREEN Hackathon 2026 on AI NAS | Track: Bring Memories to Life | Round 1 Project Proposal
Individual entry | Prepared 2026-10-27 | Deadline 2026-11-15 11:45 pm PT

Word-count note (excluded from the limit): the ten substantive answers below total 487 English words, within the 500-word limit. The third-party, licensing, team and AI-drafting section at the end is excluded from that limit under the official rules.

The ten substantive answers

1. What does the Skill do?

Story Reel turns a chosen period, event, people, or place in a family's NAS photo library into a chaptered, narrated memory album the user approves before anything is created.

2. Who is the target user and problem?

Families with years of unsorted photos on a UGREEN AI NAS. They can find single files, but turning a period like a child's summer into a shareable story means hours of manual browsing, selecting, and caption writing. Galleries show files, not memories.

3. What is the proposed solution?

Story Reel collects a small scope: a date range or event, optional people, place, tone, and output (album, document, or both). It delegates photo discovery, drafts two to five evidence-grounded chapters with the document expert, presents it, then after approval creates the album and saves the story.

4. What is the core user flow?

The user names the scope and output. Story Reel delegates photo discovery and clustering, then a factual chaptered narrative. It presents chapters, highlights, photo counts, and destinations, and waits for approval. On approval it creates the album and saves the story, then reports the real album and document.

5. What substantive difference does this offer?

Gallery search returns files; cloud album tools ship private photos to external models and often invent captions. Story Reel is evidence-grounded and approval-gated: every chapter traces to real timestamps, locations, captions, OCR, or vision results from the user's own NAS, and AI processing stays on the NAS local model.

6. Which UGREEN AI NAS capabilities are used, and at which step?

Step 1 uses ug-photo search-picture, search-classify, search-hint, meta, vision-ocr, and vision-vlm to find and cluster real photos. Step 2 uses ug-document writing for chapters. Step 4 uses ug-photo create-album or create-album-condition. Step 5 uses ug-knowledge save-text. State changes run behind gateway and workflow approval.

7. What is the technical approach?

Markdown-only orchestration delegates to photo-agent, document-agent, and knowledge-agent with delegate commands; the parent never calls cli.invoke. Inputs: scope, people, place, tone, output choice. Read-only steps gather evidence; state changes require approval. AI choices: the NAS local AI model via platform writing, OCR, and vision actions, with no external data flow. Photos stay on the NAS; the story saves only where approved.

8. What is the current development status?

Implemented: complete SKILL.md orchestration, catalog capability mapping, evidence and anti-invention rules, approval gate, state handling, and disclosures; the package passes the official checker with zero errors and warnings. Not implemented: live end-to-end expert runtime. Pending NAS validation: Action and permission behavior, approval round trips, and empty or partial results on real data.

9. What is the final validation plan?

Core case 1: a large family library, scope summer 2024; expect three to five chapters from real timestamps and places, an approved album, and a verified story document. Core case 2: Kyoto trip; expect place clustering, warm tone, and both outputs. Edge case: no matching photos or missing timestamps; expect a clear empty or partial result, no invented content, and nothing created.

10. What are the known limitations, risks, and dependencies?

Limitations: story quality depends on library metadata and AI indexes, and the parent Skill cannot see private CLI parameters. Risks: expert runtime behavior, approval latency, firmware Action availability, and large-library timeouts. Dependencies: the assigned UGREEN AI NAS iDX6011 Pro, the registered experts, and the NAS local AI model. No external services are required.

Third-party components, licensing, team information, and AI drafting disclosure (excluded from the 500-word limit)

Per the official rules, team consent declarations, member names, electronic signatures, and third-party license information are excluded from the ten-answer word limit. This section is also excluded from that limit.

Team and consent

Individual entry. No team members and no consent declarations apply. The entrant is the sole author and director of this submission.

Third-party components and platform capabilities

- Official UGOS Skill Creation Tool / Uliya Skill Kit (uliya-skill-kit), obtained from the UGREEN Hackathon Starter Kit: https://ai.ugreen.com/pages/hackathon-starter-kit. Used only as the authoring and static-validation tool. Not redistributed in the submitted ZIP.

- UGOS platform CLI capabilities used through the registered experts: ug-photo (search-picture, search-classify, search-hint, meta, vision-ocr, vision-vlm, create-album, create-album-condition), ug-document (writing), and ug-knowledge (save-text). These are UGREEN platform capabilities, not third-party libraries.
- No open-source libraries, SDKs, APIs, model weights, training datasets, or third-party data are included in this project. AI processing is designed to use the NAS local AI model only. No external services or data flows are required.

AI tools used and principal uses

An AI coding assistant was used as a drafting and review tool: to structure and edit the Markdown Skill specification, to check it against the official format, and to prepare this proposal and the documentation. Principal uses were drafting, consistency review, and static checking. The entrant directs the work, reviews every statement, decides the product behavior, and owns the submission. Consistent with the rules, the entrant remains the author and director of the Project; AI is used as a drafting tool only, and no person outside the team developed the Project or created its submission materials.

Licensing

The original text of this Project is released under the MIT License. Platform names, CLI names, and Action names belong to UGREEN and are used only to describe the supported platform capabilities.